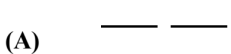
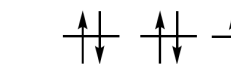
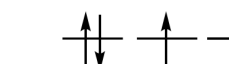
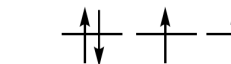
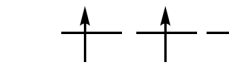


# Atomic Structure/Periodicity

- Which species has the largest ionic radius?  
(A)  $F^-$  (B)  $Na^+$  (C)  $Mg^{2+}$  (D)  $Al^{3+}$
- Which set of quantum numbers is not possible for an electron?  
(A)  $n = 1, l = 1, m_l = 0, m_s = +1/2$   
(B)  $n = 2, l = 1, m_l = 1, m_s = -1/2$   
(C)  $n = 3, l = 2, m_l = -2, m_s = +1/2$   
(D)  $n = 4, l = 0, m_l = 0, m_s = +1/2$
- For which gas-phase atoms are the  $4s$  orbitals lower in energy than the  $3d$  orbitals?  
I. H II. K  
(A) I only (B) II only  
(C) Both I and II (D) Neither I nor II
- Which element has the largest first ionization energy?  
(A) F (B) Ne (C) Cl (D) Ar
- Which best describes the distribution of electrons among the  $3d$  orbitals in  $Fe(H_2O)_6^{2+}$  in its most stable state?  
(A)  (B)   
(C)  (D)   
(A)  (B)   
(C)  (D) 
- An isotope of phosphorus undergoes  $\beta^-$  decay to give which daughter nucleus?  
(A)  $^{27}Al$  (B)  $^{29}Si$  (C)  $^{31}P$  (D)  $^{33}S$
- Which atom has the smallest first ionization energy?  
(A) Mg (B) Al (C) Si (D) P
- How many electrons in a ground-state atom of uranium (U) have the principal quantum number  $n = 4$ ?  
(A) 4 (B) 16 (C) 18 (D) 32
- Which property of a photon is directly proportional to its frequency?  
(A) Energy (B) Momentum  
(C) Velocity (D) Wavelength
- Photoelectron spectroscopy of Ar provides information about which properties of Ar?  
I. Ionization energy II. Electronic subshell structure  
(A) I only (B) II only  
(C) Both I and II (D) Neither I nor II
- Which statements about fluorine are correct?  
I. It has the highest electronegativity of any element.  
II. Its electron affinity is the largest in magnitude of any element.  
(A) I only (B) II only  
(C) Both I and II (D) Neither I nor II
- Thorium-232 decays by emission first of an alpha particle, then of a beta particle, then of a second beta particle, and then of a second alpha particle. What is the daughter nucleus after the fourth disintegration?  
(A)  $^{228}Th$  (B)  $^{228}Ac$   
(C)  $^{224}Ra$  (D)  $^{224}Rn$
- Which element's electronegativity is most similar to that of sulfur (S)?  
(A) O (B) F (C) As (D) Se
- Two atoms of the same element, A and B, are in identical excited states. Atom A emits a photon of red light, while atom B emits a photon of blue light. Which statement best describes the energies of the two atoms after the emission of light?  
(A) Atom A is in a lower-energy state than atom B.  
(B) Atom A is in a higher-energy state than atom B.  
(C) Atom A and atom B have the same energy.  
(D) The relative energies of A and B cannot be determined from the information given.
- Two orbitals in an atom are both spherically symmetric but differ in size. Which statements about the orbitals' quantum numbers must be correct?  
I. The two orbitals have different values of  $n$ .  
II. The two orbitals have the same value of  $m_l$ .  
(A) I only (B) II only  
(C) Both I and II (D) Neither I nor II

# Atomic Structure/Periodicity

16. Which is a pair of gas-phase species with the same ground state electron configuration?
- (A) V and  $\text{Cr}^+$  (B)  $\text{Mn}^+$  and  $\text{Fe}^-$   
(C) Ni and Pd (D) Zn and  $\text{Ga}^+$
17. Which list arranges from highest to lowest the standard reduction potentials  $E^\circ$  for reduction of the divalent group 10 metal ions  $\text{M}^{2+}(\text{aq})$  to  $\text{M}(\text{s})$ ?
- (A)  $\text{Ni} > \text{Pd} > \text{Pt}$  (B)  $\text{Pt} > \text{Pd} > \text{Ni}$   
(C)  $\text{Pd} > \text{Ni} > \text{Pt}$  (D)  $\text{Pd} > \text{Pt} > \text{Ni}$
18. A nucleus absorbs a neutron and undergoes fission to produce  $^{103}_{40}\text{Zr}$ ,  $^{134}_{54}\text{Xe}$ , and three additional neutrons. What is the fissile nucleus?
- (A)  $^{235}\text{U}$  (B)  $^{237}\text{U}$  (C)  $^{237}\text{Pu}$  (D)  $^{239}\text{Pu}$
19. Which species has the largest ionic radius?
- (A)  $\text{S}^{2-}$  (B)  $\text{Cl}^-$  (C)  $\text{K}^+$  (D)  $\text{Ca}^{2+}$
20. Which of the following statements regarding ionization energy of gas-phase atoms is correct?
- (A) The first ionization energy of a group 1 element is always greater than the first ionization energy of the group 2 element in the same row of the periodic table.  
(B) In the second row of the periodic table, the first ionization energy is directly proportional to the atomic radius.  
(C) Among the elements from Al ( $Z = 13$ ) to Ar ( $Z = 18$ ), the first ionization energy increases monotonically with atomic number.  
(D) Among the elements of group 16, the first ionization energy decreases monotonically with atomic number.
21. How many electrons in a ground state gas-phase Al atom have the quantum number  $l = 1$ ?
- (A) 1 (B) 3 (C) 5 (D) 7
22. What is the ground-state electron configuration of a gas-phase  $\text{Mn}^{2+}$  ion?
- (A)  $1s^2 2s^2 2p^6 3s^2 3p^6 3d^5$  (B)  $1s^2 2s^2 2p^6 3s^2 3p^6 3d^3 4s^2$   
(C)  $1s^2 2s^2 2p^6 3s^2 3p^6 3d^7$  (D)  $1s^2 2s^2 2p^6 3s^2 3p^6 3d^5 4s^2$
23. What wavelengths of light are capable of breaking a bond with a bond strength of  $425 \text{ kJ mol}^{-1}$ ?
- (A) Only  $\lambda < 281 \text{ nm}$  (B) Only  $\lambda > 281 \text{ nm}$   
(C) Only  $\lambda < 355 \text{ nm}$  (D) Only  $\lambda > 355 \text{ nm}$
24. When  $^{105}\text{Te}$  decays by alpha emission, how many protons are in the daughter nuclide that is produced?
- (A) 48 (B) 50 (C) 52 (D) 53
25. Which statement best describes the amount of energy released by gas-phase oxygen and sulfur atoms upon accepting an electron (electron affinities)?
- (A) O releases less energy than S because the smaller size of O causes greater electron-electron repulsion.  
(B) O releases less energy than S because O has a lower mass than S.  
(C) O releases more energy than S because the electron enters a  $2p$  orbital in O but a  $3p$  orbital in S.  
(D) O releases more energy than S because O has a larger effective nuclear charge  $Z_{\text{eff}}$  than S.
26. Which element has chemical properties most similar to those of S?
- (A) O (B) P (C) Cl (D) Se
27. What is the largest number of electrons that can have both a principal quantum number  $n = 4$  and an orbital angular momentum quantum number  $l = 2$  in a given atom?
- (A) 1 (B) 3 (C) 5 (D) 10
28. Which best represents the arrangement of electrons in a ground-state gas-phase  $\text{Ni}^{2+}$  ion?
- $$\begin{array}{c} 4s \qquad \qquad 3d \\ \uparrow \qquad \qquad \uparrow \uparrow \uparrow \uparrow \uparrow \uparrow \end{array}$$
- (A)  $[\text{Ar}] (\uparrow\downarrow) (\uparrow\downarrow)(\uparrow\downarrow)(\uparrow\downarrow)(\ )(\ )$   
(B)  $[\text{Ar}] (\uparrow\downarrow) (\uparrow\downarrow)(\uparrow)(\uparrow)(\uparrow)(\uparrow)$   
(C)  $[\text{Ar}] (\ ) (\uparrow\downarrow)(\uparrow\downarrow)(\uparrow\downarrow)(\uparrow\downarrow)(\ )$   
(D)  $[\text{Ar}] (\ ) (\uparrow\downarrow)(\uparrow\downarrow)(\uparrow\downarrow)(\uparrow)(\uparrow)$
29. What is the wavelength of a photon whose energy is  $334 \text{ kJ mol}^{-1}$ ?
- (A) 222 nm (B) 358 nm  
(C) 590 nm (D) 2994 nm

# Atomic Structure/Periodicity

30. What nuclide decays by alpha emission to form  $^{219}\text{Rn}$ ?  
 (A)  $^{215}\text{Po}$  (B)  $^{219}\text{At}$  (C)  $^{219}\text{Fr}$  (D)  $^{223}\text{Ra}$
31. Photons of what wavelength of light have an energy of  $1.00 \times 10^3 \text{ kJ mol}^{-1}$ ?  
 (A)  $2.00 \times 10^{31} \text{ m}$  (B)  $1.20 \times 10^{-7} \text{ m}$   
 (C)  $1.20 \times 10^{-4} \text{ m}$  (D)  $8.36 \times 10^6 \text{ m}$
32. In which set are all three species isoelectronic?  
 (A)  $\text{H}^+$ ,  $\text{Li}^+$ ,  $\text{Na}^+$  (B)  $\text{C}$ ,  $\text{N}^-$ ,  $\text{O}^{2-}$   
 (C)  $\text{Se}^{2-}$ ,  $\text{Br}^-$ ,  $\text{Rb}^+$  (D)  $\text{I}$ ,  $\text{Xe}$ ,  $\text{Cs}$
33. Which change in principal quantum number  $n$  involves the greatest change in energy?  
 (A)  $n = 1 \rightarrow n = 2$  in  $\text{H}$  (B)  $n = 2 \rightarrow n = 3$  in  $\text{He}^+$   
 (C)  $n = 3 \rightarrow n = 4$  in  $\text{Li}^{2+}$  (D)  $n = 4 \rightarrow n = 5$  in  $\text{Be}^{3+}$
34. Which gas-phase ion has the greatest number of unpaired electrons in its ground state?  
 (A)  $\text{Fe}^{3+}$  (B)  $\text{Co}^{3+}$  (C)  $\text{Ni}^{3+}$  (D)  $\text{Cu}^{3+}$
35. Which principle restricts the occupancy of an atomic orbital to no more than two electrons?  
 (A) The principle of conservation of energy  
 (B) The principle of conservation of angular momentum  
 (C) The Pauli exclusion principle  
 (D) The Heisenberg uncertainty principle
36. When copper-64 decays by positron emission, what daughter nuclide is formed?  
 (A)  $^{64}\text{Ni}$  (B)  $^{65}\text{Ni}$  (C)  $^{64}\text{Zn}$  (D)  $^{65}\text{Zn}$
37. Which is a possible set of quantum numbers  $n$ ,  $l$ ,  $m_l$ ,  $m_s$  for a valence electron of sulfur (S)?  
 (A) 3, 2, 2,  $\frac{1}{2}$  (B) 3, 1, -1,  $\frac{1}{2}$   
 (C) 2, 1, 0,  $\frac{1}{2}$  (D) 3, 2, 0,  $-\frac{1}{2}$
38. Which period 3 element has these successive ionization energies (in  $\text{kJ mol}^{-1}$ )?

| $IE_1$ | $IE_2$ | $IE_3$ | $IE_4$ |
|--------|--------|--------|--------|
| 786.5  | 1577.1 | 3231.6 | 4355.5 |
| $IE_5$ | $IE_6$ | $IE_7$ |        |
| 16091  | 19805  | 23780  |        |

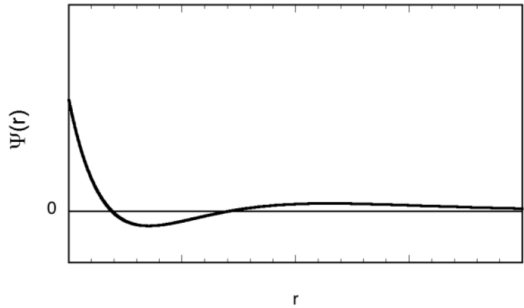
- (A) Al (B) Si (C) P (D) S

39. Which is the ground-state electron configuration of gas-phase  $\text{Co}^{2+}$ ?  
 (A)  $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^7$   
 (B)  $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^5$   
 (C)  $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 4d^5$   
 (D)  $1s^2 2s^2 2p^6 3s^2 3p^6 3d^7$
40. A fluorescent dye absorbs a photon of light of 485 nm and emits a photon of light at 540 nm. How much energy is lost as heat?  
 (A)  $4.17 \times 10^{-20} \text{ J}$  (B)  $3.68 \times 10^{-19} \text{ J}$   
 (C)  $4.10 \times 10^{-19} \text{ J}$  (D)  $3.06 \times 10^{-18} \text{ J}$
41. Which gas-phase atom releases the most energy when an electron is added to it?  
 (A) Na (B) Cl (C) K (D) Br
42. The isotope  $^{44}\text{Ti}$  undergoes electron capture. Which daughter isotope is produced?  
 (A)  $^{40}\text{K}$  (B)  $^{40}\text{Ca}$  (C)  $^{44}\text{Sc}$  (D)  $^{44}\text{V}$
43. For which element is the +2 oxidation state LEAST common?  
 (A) Al (B) V (C) Fe (D) Sm
44. How many completely filled  $p$  orbitals are present in a ground state Cl atom?  
 (A) 1 (B) 2 (C) 4 (D) 5
45. Which atom has the highest first ionization energy?  
 (A) N (B) O (C) P (D) S
46. The electronegativity of gallium (1.8) is greater than that of aluminum (1.6). Which is the best explanation for this difference?  
 (A) Ga is larger than Al, so its valence electrons experience less electron-electron repulsion.  
 (B) Ga has more protons than Al, so its valence electrons are more attracted to the nucleus.  
 (C) Ga has a filled  $3d$  subshell, whose electrons incompletely screen the nucleus.  
 (D) Ga experiences significant relativistic effects, which contract the valence orbitals.

# Atomic Structure/Periodicity

47. Which set of quantum numbers could NOT correspond to an electron in a ground-state gas phase Pd atom?
- (A)  $n = 2, l = 1, m_l = -1, m_s = 1/2$   
(B)  $n = 3, l = 3, m_l = -1, m_s = -1/2$   
(C)  $n = 4, l = 0, m_l = 0, m_s = -1/2$   
(D)  $n = 4, l = 2, m_l = 2, m_s = 1/2$
48. A certain nuclide undergoes beta decay to form  $^{70}\text{Ge}$ . If the same nuclide undergoes electron capture instead, what daughter nuclide forms?
- (A)  $^{70}\text{Zn}$  (B)  $^{70}\text{Ga}$  (C)  $^{70}\text{Ge}$  (D)  $^{70}\text{As}$
49. Which statement regarding the relative energies of monochromatic light with  $\lambda = 800 \text{ nm}$  and monochromatic light with  $\lambda = 400 \text{ nm}$  is correct?
- (A) 800 nm light has half as much energy per mole of photons as 400 nm light.  
(B) 800 nm light has the same energy per mole of photons as 400 nm light.  
(C) 800 nm light has twice as much energy per mole of photons as 400 nm light.  
(D) No conclusion may be drawn regarding the relative energy per mole of photons without knowing the intensity of the light.
50. How many electrons in a ground-state As atom in the gas phase have quantum numbers  $n = 3$  and  $l = 1$ ?
- (A) 1 (B) 2 (C) 3 (D) 6
51. In which list are the elements in order of increasing first ionization energy?
- (A)  $\text{Ca} < \text{Si} < \text{P} < \text{N}$  (B)  $\text{N} < \text{P} < \text{Si} < \text{Ca}$   
(C)  $\text{Ca} < \text{N} < \text{P} < \text{Si}$  (D)  $\text{N} < \text{Si} < \text{P} < \text{Ca}$
52. What mode of radioactive decay does  $^{66}\text{Cu}$  undergo?
- (A) Alpha decay (B) Beta decay  
(C) Electron capture (D) Positron emission
53. Which gas-phase atoms in their ground states are diamagnetic?
- I. Fe  
II. Zn
- (A) I only (B) II only  
(C) Both I and II (D) Neither I nor II
54. The boiling points of the alkali metals decrease from lithium to cesium. Which trend is most closely related to the decreasing boiling points?
- (A) Increasing atomic radius down the group  
(B) Increasing atomic mass down the group  
(C) Decreasing electronegativity down the group  
(D) Decreasing electron affinity down the group
55. What is the electron configuration of the  $\text{Al}^{3+}$  ion?
- (A)  $1s^2 2s^2 2p^6$  (B)  $1s^2 2s^2 2p^6 3s^2 3p^1$   
(C)  $1s^2 2s^2 2p^6 3s^2 3p^4$  (D)  $1s^2 2s^2 2p^6 3s^2 3p^6$
56. Which metal has the highest melting point?
- (A) K (B) Ca (C) Fe (D) Zn
57. The wavelength of one line in the emission spectrum of C is 538 nm. What is the energy of one photon with this wavelength?
- (A)  $3.69 \times 10^{-19} \text{ J}$  (B)  $3.69 \times 10^{-26} \text{ J}$   
(C)  $3.56 \times 10^{-40} \text{ J}$  (D)  $1.19 \times 10^{-48} \text{ J}$
58. An orbital in a ground-state gas-phase As atom has  $n = 3$ ,  $l = 1$ . How many electrons are in this orbital?
- (A) 0 (B) 1 (C) 2 (D) 6
59. Atoms of which element have the smallest ionization energy?
- (A) Na (B) F (C) K (D) Cl
60. The acidity of the group 16 hydrides increase going down the group ( $\text{H}_2\text{O} \ll \text{H}_2\text{S} < \text{H}_2\text{Se} < \text{H}_2\text{Te}$ ). Which is the best explanation for this trend?
- (A) The electronegativity of the group 16 elements increases going down the group.  
(B) The polarizability of the group 16 elements increases going down the group.  
(C) The polarity of the X–H bond increases going down the group.  
(D) The H–X–H bond angle increases going down the group.
61. How many total orbitals are there with principal quantum number  $n = 4$ ?
- (A) 1 (B) 4 (C) 9 (D) 16

# Atomic Structure/Periodicity

62. Addition of an electron to a gas-phase Si atom results in the release of more energy than addition of an electron to a gas-phase P atom. What is the best explanation for their relative electron affinities?
- (A) The electron added to Si experiences less electron-electron repulsion than the electron added to P.  
(B) The electron added to Si enters a lower-energy subshell than the electron added to P.  
(C) Si is more electronegative than P.  
(D) Si is smaller than P.
63. The wavelength of one of the spectral lines of helium is 492 nm. What is the energy of a photon with this wavelength?
- (A)  $3.26 \times 10^{-40}$  J      (B)  $3.26 \times 10^{-31}$  J  
(C)  $4.04 \times 10^{-28}$  J      (D)  $4.04 \times 10^{-19}$  J
64. Which of the following gas-phase ions has the largest number of unpaired electrons in its ground state?
- (A)  $\text{Cr}^{3+}$       (B)  $\text{Co}^{3+}$       (C)  $\text{Ni}^{2+}$       (D)  $\text{Cu}^{2+}$
65.  $^{52}\text{Cr}$  undergoes radioactive decay to give  $^{52}\text{Cr}$  by what decay mode?
- (A) Alpha emission      (B) Beta emission  
(C) Positron emission      (D) Gamma emission
66. An orbital has the radial wavefunction shown below. What orbital is it?
- 
- (A) 1s      (B) 2s      (C) 3s      (D) 4s
67. Which of the following statements is best supported by the data from Rutherford's experiment of scattering alpha particles with a thin metal foil?
- (A) The mass and positive charge of an atom are concentrated in its center.  
(B) Electrons in atoms occupy only certain specific energy levels.  
(C) Moving particles can also be described as waves.  
(D) Atoms of a given element do not all have identical masses.
68. What is the formula of the most stable oxide of francium?
- (A)  $\text{Fr}_2\text{O}$       (B)  $\text{FrO}$       (C)  $\text{Fr}_2\text{O}_3$       (D)  $\text{FrO}_2$
69. As atomic number increases from 11 to 15, the atomic radii of the elements
- (A) increase  
(B) decrease  
(C) increase, then decrease  
(D) decrease, then increase
70. Nitrogen has a greater first ionization energy than oxygen. What is the best explanation for this observation?
- (A) Nitrogen is more electronegative than oxygen.  
(B) A nitrogen atom is smaller than an oxygen atom.  
(C) The electron ionized from nitrogen experiences less electron-electron repulsion than the electron ionized from oxygen.  
(D) The electron ionized from nitrogen is a 2s electron, while the electron ionized from oxygen is a 2p electron.
71. Which set of quantum numbers  $n, l, m_l, m_s$  is invalid?
- (A) 1, 1, 0,  $-\frac{1}{2}$       (B) 2, 0, 0,  $+\frac{1}{2}$   
(C) 3, 1, 0,  $+\frac{1}{2}$       (D) 4, 3, 2,  $-\frac{1}{2}$
72. What is the product of alpha emission from the isotope uranium-238?
- (A)  $^{232}\text{Th}$       (B)  $^{234}\text{Th}$       (C)  $^{237}\text{Np}$       (D)  $^{231}\text{Pa}$